

# Context-Free Grammar (CFG)

CFG stands for context-free grammar. It is a formal grammar which is used to generate all possible patterns of strings in a given formal language. Context-free grammar  $G$  can be defined by four tuples as:

1.  $G = (V, T, P, S)$

**Where,**

$G$  is the grammar, which consists of a set of the production rule. It is used to generate the string of a language.

$T$  is the final set of a terminal symbol. It is denoted by lower case letters.

$V$  is the final set of a non-terminal symbol. It is denoted by capital letters.

$P$  is a set of production rules, which is used for replacing non-terminals symbols(on the left side of the production) in a string with other terminal or non-terminal symbols(on the right side of the production).

$S$  is the start symbol which is used to derive the string. We can derive the string by repeatedly replacing a non-terminal by the right-hand side of the production until all non-terminal have been replaced by terminal symbols.

## Example 1:

Construct the CFG for the language having any number of a's over the set  $\Sigma = \{a\}$ .

**Solution:**

As we know the regular expression for the above language is

r.e. =  $a^*$

Production rule for the Regular expression is as follows:

$S \rightarrow aS$       rule 1

$S \rightarrow \epsilon$       rule 2

Now if we want to derive a string "aaaaaa", we can start with start symbols.

S  
aS  
aaS            rule 1  
aaaS           rule 1  
aaaaS          rule 1  
aaaaaS        rule 1  
aaaaaaS       rule 1  
aaaaaaε       rule 2  
aaaaaa

The r.e. =  $a^*$  can generate a set of string  $\{\epsilon, a, aa, aaa, \dots\}$ . We can have a null string because S is a start symbol and rule 2 gives  $S \rightarrow \epsilon$ .

### Example 2:

Construct a CFG for the regular expression  $(0+1)^*$

#### **Solution:**

The CFG can be given by,

Production rule (P):

$S \rightarrow 0S \mid 1S$

$S \rightarrow \epsilon$

The rules are in the combination of 0's and 1's with the start symbol. Since  $(0+1)^*$  indicates  $\{\epsilon, 0, 1, 01, 10, 00, 11, \dots\}$ . In this set,  $\epsilon$  is a string, so in the rule, we can set the rule  $S \rightarrow \epsilon$ .

### Example 3:

Construct a CFG for a language  $L = \{wcwR \mid \text{where } w \in (a, b)^*\}$ .

#### **Solution:**

The string that can be generated for a given language is  $\{aaciaa, bcb, abcba, bacab, abbcbb, \dots\}$

The grammar could be:

$S \rightarrow aSa$       rule 1  
 $S \rightarrow bSb$       rule 2  
 $S \rightarrow c$       rule 3

Now if we want to derive a string "abbcbbba", we can start with start symbols.

$S \rightarrow aSa$   
 $S \rightarrow abSba$       from rule 2  
 $S \rightarrow abbSbba$       from rule 2  
 $S \rightarrow abbcbbba$       from rule 3

Thus any of this kind of string can be derived from the given production rules.

### Example 4:

Construct a CFG for the language  $L = a^n b^{2n}$  where  $n \geq 1$ .

#### **Solution:**

The string that can be generated for a given language is {abb, aabbbb, aaabbbbbbb. }.

The grammar could be:

$S \rightarrow aSbb \mid abb$

Now if we want to derive a string "aabbbb", we can start with start symbols.

$S \rightarrow aSbb$   
 $S \rightarrow aabbbb$

# Derivation

Derivation is a sequence of production rules. It is used to get the input string through these production rules. During parsing, we have to take two decisions. These are as follows:

- We have to decide the non-terminal which is to be replaced.
- We have to decide the production rule by which the non-terminal will be replaced.

We have two options to decide which non-terminal to be placed with production rule.

## 1. Leftmost Derivation:

In the leftmost derivation, the input is scanned and replaced with the production rule from left to right. So in leftmost derivation, we read the input string from left to right.

### Example:

#### Production rules:

$$E = E + E$$

$$E = E - E$$

$$E = a \mid b$$

#### Input

$$a - b + a$$

#### The leftmost derivation is:

$$E = E + E$$

$$E = E - E + E$$

$$E = a - E + E$$

$$E = a - b + E$$

$$E = a - b + a$$

## 2. Rightmost Derivation:

In rightmost derivation, the input is scanned and replaced with the production rule from right to left. So in rightmost derivation, we read the input string from right to left.

## Example

### Production rules:

$$E = E + E$$

$$E = E - E$$

$$E = a \mid b$$

### Input

$$a - b + a$$

### The rightmost derivation is:

$$E = E - E$$

$$E = E - E + E$$

$$E = E - E + a$$

$$E = E - b + a$$

$$E = a - b + a$$

When we use the leftmost derivation or rightmost derivation, we may get the same string. This type of derivation does not effect on getting of a string.

## Examples of Derivation:

### Example 1:

Derive the string "abb" for leftmost derivation and rightmost derivation using a CFG given by,

$$S \rightarrow AB \mid \epsilon$$

$$A \rightarrow aB$$

$$B \rightarrow Sb$$

### Solution:

#### Leftmost derivation:

S

AB

aB B

a Sb B

A  $\varepsilon$  bB

ab Sb

ab  $\varepsilon$  b

abb

### Rightmost derivation:

S

AB

A Sb

A  $\varepsilon$  b

aB b

a Sb b

a  $\varepsilon$  bb

abb

### Example 2:

Derive the string "aabbabba" for leftmost derivation and rightmost derivation using a CFG given by,

$S \rightarrow aB \mid bA$

$S \rightarrow a \mid aS \mid bAA$

$S \rightarrow b \mid aS \mid aBB$

### Solution:

### Leftmost derivation:

$S$   
 $aB \quad S \rightarrow aB$   
 $aaBB \quad B \rightarrow aBB$   
 $aabB \quad B \rightarrow b$   
 $aabbS \quad B \rightarrow bS$   
 $aabbaB \quad S \rightarrow aB$   
 $aabbabS \quad B \rightarrow bS$   
 $aabbabbA \quad S \rightarrow bA$   
 $aabbabba \quad A \rightarrow a$

### Rightmost derivation:

$S$   
 $aB \quad S \rightarrow aB$   
 $aaBB \quad B \rightarrow aBB$   
 $aaBbS \quad B \rightarrow bS$   
 $aaBbbA \quad S \rightarrow bA$   
 $aaBbba \quad A \rightarrow a$   
 $aabSbba \quad B \rightarrow bS$   
 $aabbAbba \quad S \rightarrow bA$   
 $aabbabba \quad A \rightarrow a$

### Example 3:

Derive the string "00101" for leftmost derivation and rightmost derivation using a CFG given by,

$S \rightarrow A1B$   
 $A \rightarrow 0A \mid \epsilon$   
 $B \rightarrow 0B \mid 1B \mid \epsilon$

### Solution:

#### Leftmost derivation:

$S$   
 $A1B$   
 $0A1B$   
 $00A1B$   
 $001B$   
 $0010B$

00101B

00101

**Rightmost derivation:**

S

A1B

A10B

A101B

A101

0A101

00A101

00101